Introduction:

This laboratory assignment presents the complete design flow of a CMOS digital circuit using the Cadence Virtuoso design environment. The project focuses on implementing the required CMOS circuit using the gpdk045 (45 nm) technology within a custom library named lib7. The work covers the essential stages of custom integrated circuit (IC) design, including schematic creation, symbol generation, pre-layout circuit simulation, physical layout implementation, and comprehensive design verification.

Throughout the experiment, industry-standard Electronic Design Automation (EDA) tools were used to transform the circuit from a transistor-level schematic into a manufacturable layout. A dedicated testbench was developed to evaluate the circuit through both DC and transient analyses, allowing its static and dynamic performance to be examined.

To ensure both functional and physical correctness, several verification processes were performed, including Design Rule Check (DRC) to verify compliance with fabrication rules, Layout Versus Schematic (LVS) to confirm consistency between the schematic and layout, and Parasitic Extraction to include the effects of parasitic resistances and capacitances in post-layout simulations.

Finally, the simulation results obtained before and after parasitic extraction were compared to evaluate the influence of the physical layout on circuit performance. This experiment provides practical experience with the complete CMOS design flow and demonstrates the importance of verification, layout optimization, and parasitic analysis in modern VLSI design.

Objectives

The main objectives of this laboratory assignment are:

- To design and implement a CMOS digital circuit using the Cadence Virtuoso design environment.
- To create a transistor-level schematic and verify its correctness using the Check and Save function.
- To generate a reusable symbol for the designed circuit and integrate it into a simulation testbench.
- To perform both DC and Transient simulations to evaluate the electrical performance of the circuit.

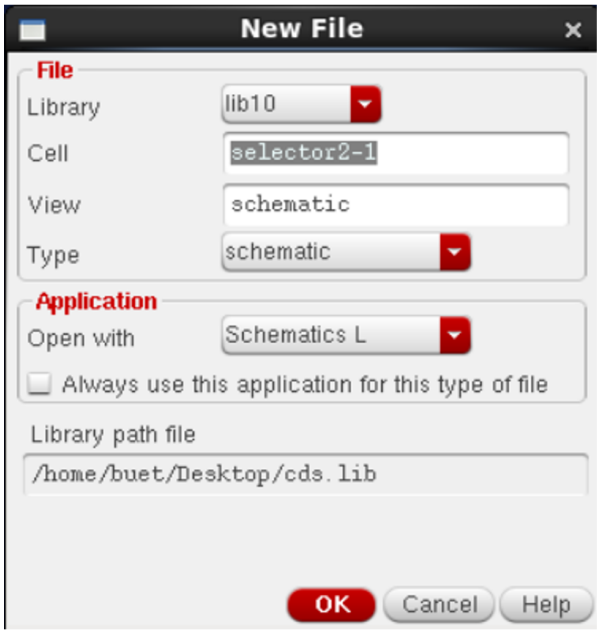
- To develop the physical layout while following the design rules of the gpdk045 technology.
- To verify the layout using Design Rule Check (DRC) and Layout Versus Schematic (LVS).
- To perform Parasitic Extraction and analyze the impact of parasitic effects through post-layout simulation.
- To gain practical experience with the complete CMOS IC design flow used in modern VLSI design.

Schematic Design in Cadence Virtuoso

Step 1: Create a New Schematic Cell View

The Cadence Virtuoso environment was launched, and a new library named **lib7** was created and attached to the **gpdk045** technology library. A new **Schematic Cell View** was then generated to begin the circuit design.

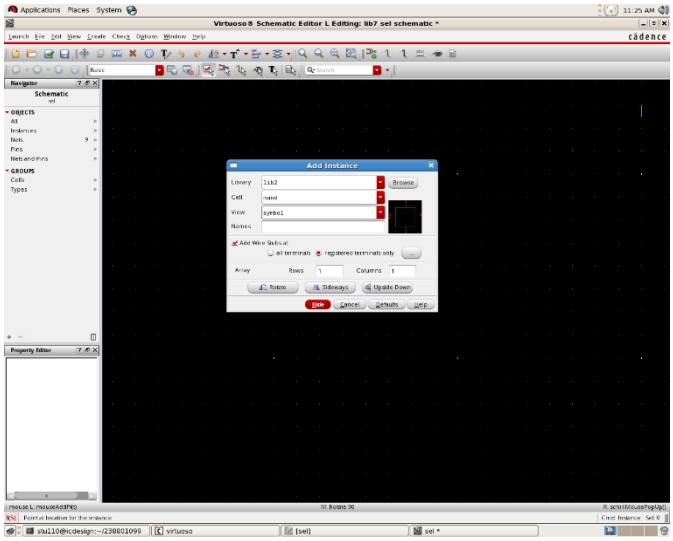
The required components were selected from the **gpdk045** library, including the PMOS and NMOS transistors, power supply (**VDD**), ground (**GND**), and the required input and output pins for the designed circuit.

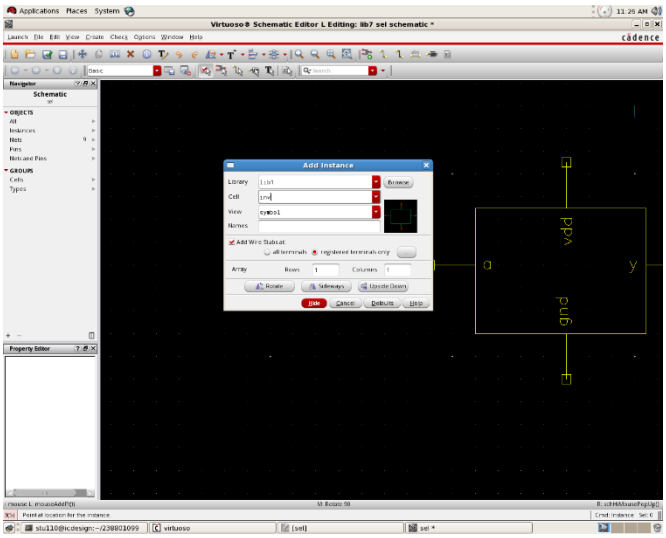


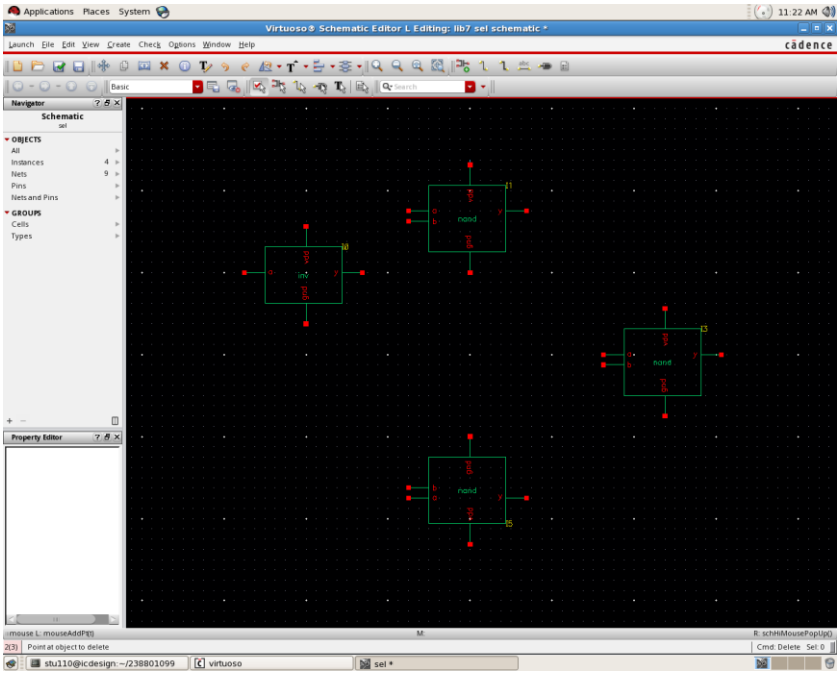
Step 2: Choose and Place Components

After creating the schematic in lib7, the required circuit blocks were inserted using the Add Instance tool. Previously designed cells were selected from different libraries to build the complete circuit. In this project, the NAND cell was imported from lib2, while the Inverter (inv) cell was imported from lib1. Both cells were inserted into the lib7 schematic using their symbol views.

This hierarchical design approach allows previously designed and verified circuit blocks to be reused, reducing design time and improving design reliability. Each component was selected by specifying its library, cell, and view before being placed in the schematic edito



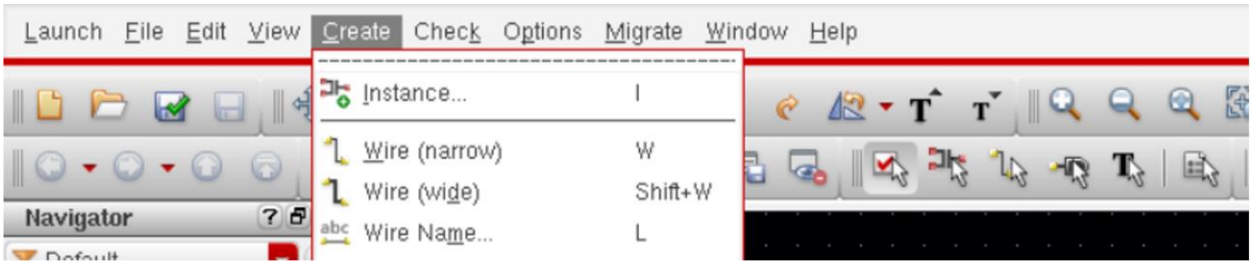




Step 3: Wire the Circuit

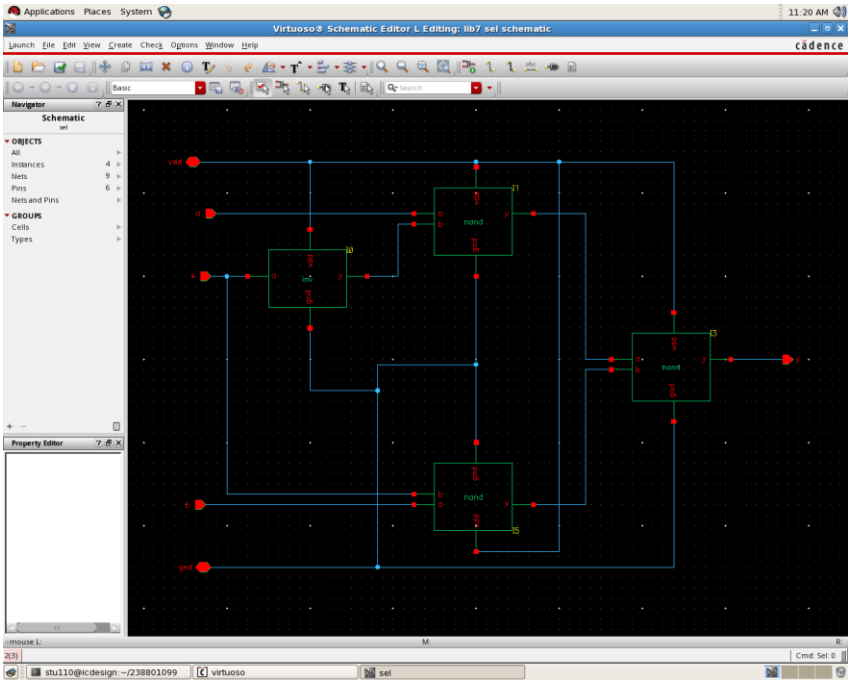
The inserted circuit blocks were connected using the Wire tool in the Cadence Schematic Editor. Proper electrical connections were established between the input and output terminals, as well as the power rails (VDD and GND), according to the required circuit design.

After completing the wiring, all signal paths were carefully checked to ensure correct connectivity and proper circuit operation before proceeding to the verification stage.



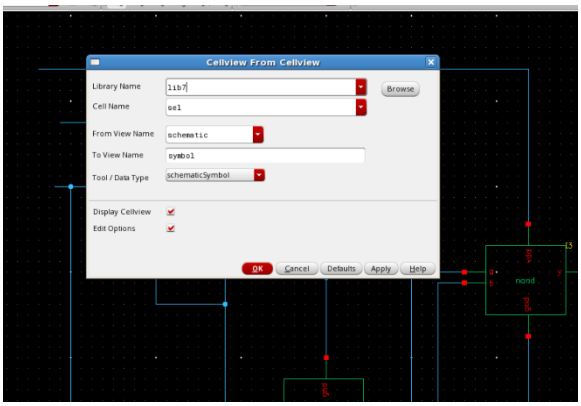
Step 4: Check and Save the Schematic

After completing the schematic, the Check and Save function was executed to verify the design. The verification process confirmed that there were no connectivity or electrical errors, indicating that the schematic was correctly implemented and ready for the next stages of the design flow, including symbol generation and circuit simulatio



Symbol Creation

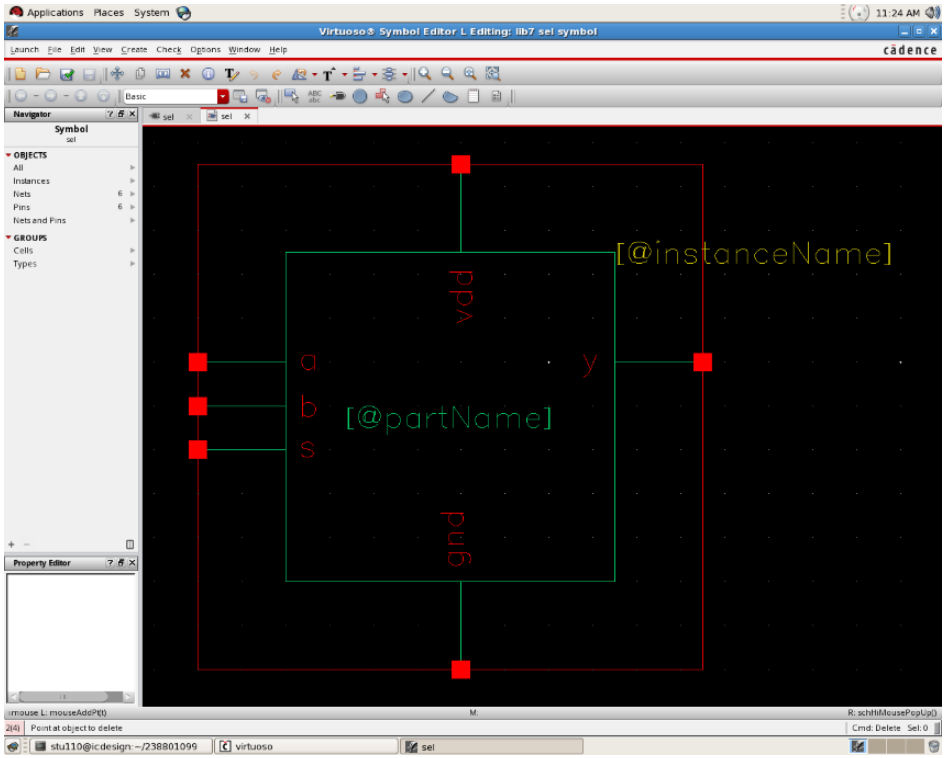
After successfully completing the schematic design and verifying it using the **Check and Save** function, a **symbol** was generated from the schematic using the **Create → Cellview → From Cellview** command in Cadence Virtuoso.



The generated symbol provides a simplified graphical representation of the designed circuit, allowing it to be reused in higher-level designs without exposing the transistor-level implementation. During the symbol generation process, the input and output terminals were automatically imported from the schematic and arranged to provide a clear and organized interface.

Using symbols is an important feature of hierarchical design methodology because it improves circuit organization, enhances design reusability, and simplifies the construction of larger digital systems and simulation testbenches.

The generated symbol was verified to ensure that all pins were correctly assigned and that it accurately represented the original schematic before proceeding to the simulation stage.

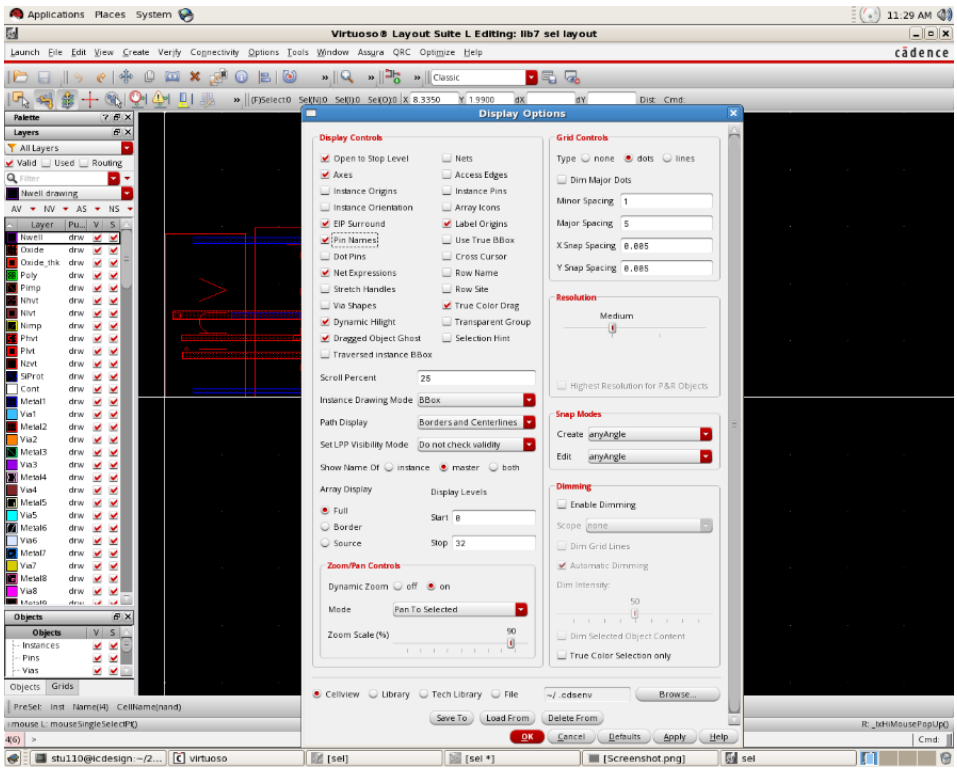


Layout Design

After successfully completing and verifying the schematic design, the physical layout of the circuit was created using the **Cadence Virtuoso Layout Editor**. The layout represents the physical implementation of the designed CMOS circuit and was developed in accordance with the design rules defined by the **gpdk045** technology.

Step 1: Configure the Layout Environment

Before starting the layout implementation, the **Display Options** window was configured to optimize the layout workspace. The display settings were adjusted to improve the visibility of layout layers, routing paths, pin names, and the design grid. These configurations provide a clearer working environment and help ensure accurate placement and routing throughout the layout design process.



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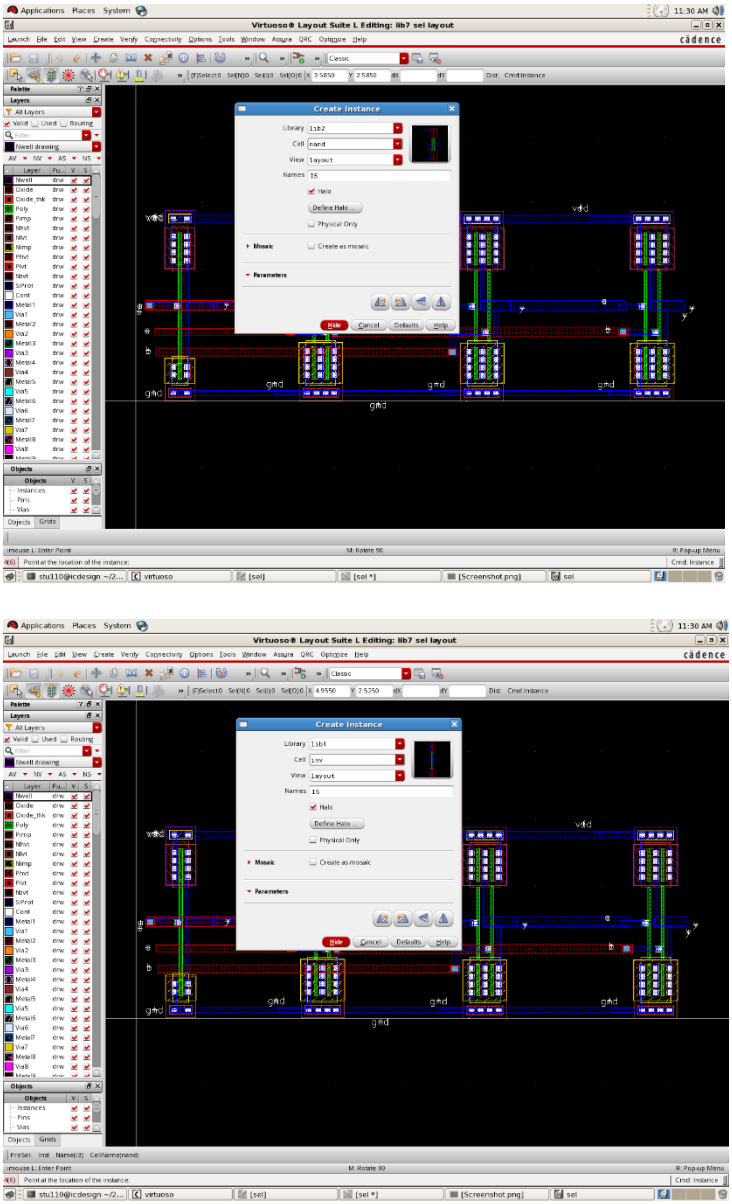
Step 2: Place the Layout Cells

After configuring the layout environment, the required layout cells were inserted into the **lib7** layout using the **Create Instance** tool. A hierarchical design approach was adopted by reusing previously created layout cells from different libraries.

In this design, **three NAND layout cells** were imported from **lib2**, while **one Inverter (INV) layout cell** was imported from **lib1**. Each layout cell was selected through the **Create Instance** window and positioned according to the required circuit structure.

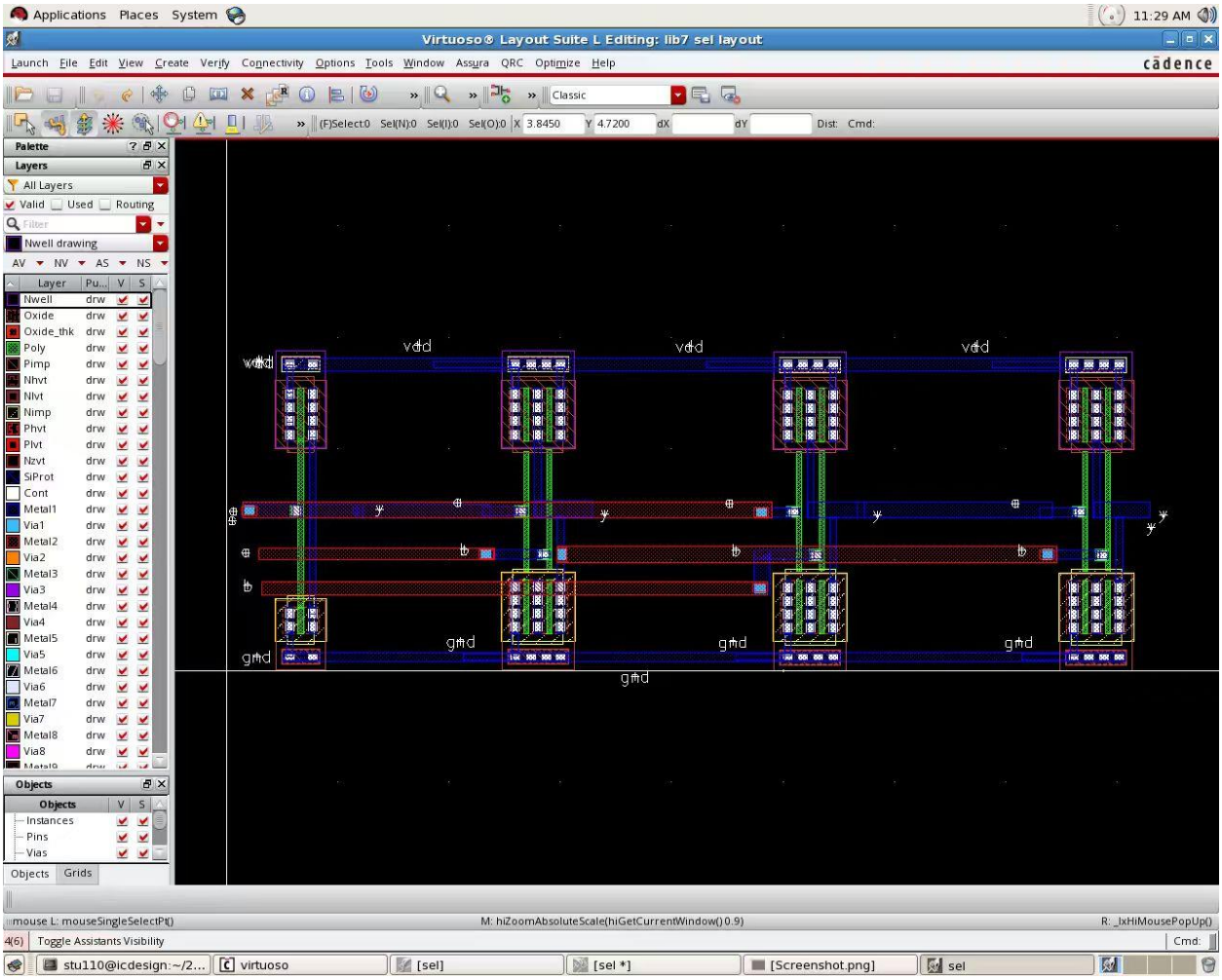
This hierarchical methodology improves design efficiency, promotes circuit reusability, and simplifies the implementation of larger CMOS circuits while maintaining consistency with previously verified layout cells.

Figure 8. Adding the **INV** layout cell from **lib1** using the **Create Instance** tool.



Final Layout

After placing all required cells and completing the routing, the final physical layout of the **SEL** circuit was obtained. The layout was constructed hierarchically by instantiating **three NAND gate layouts from lib2** and **one inverter layout from lib1** inside **lib7**. All signal connections, together with the **VDD** and **GND** power rails, were completed using metal layers while following the **gpdk045** layout design rules. The completed layout



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DRC Results

After running the DRC verification, the results showed that **no design rule violations** were found in the layout. The circuit successfully passed all design rule checks, confirming that the physical layout complies with the fabrication constraints of the **gpdk045 (45 nm)** technology.

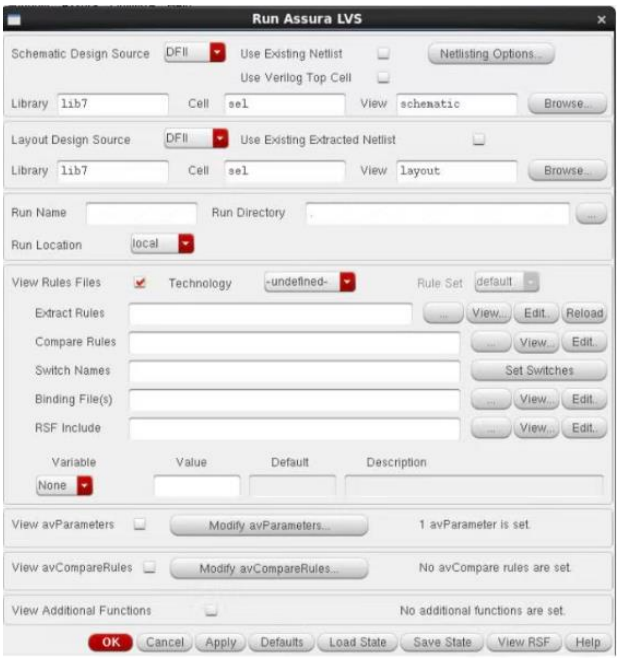
This successful verification indicates that the layout is ready for the next verification stage.



Running LVS

After successfully completing the DRC verification, the **Layout Versus Schematic (LVS)** verification was performed.

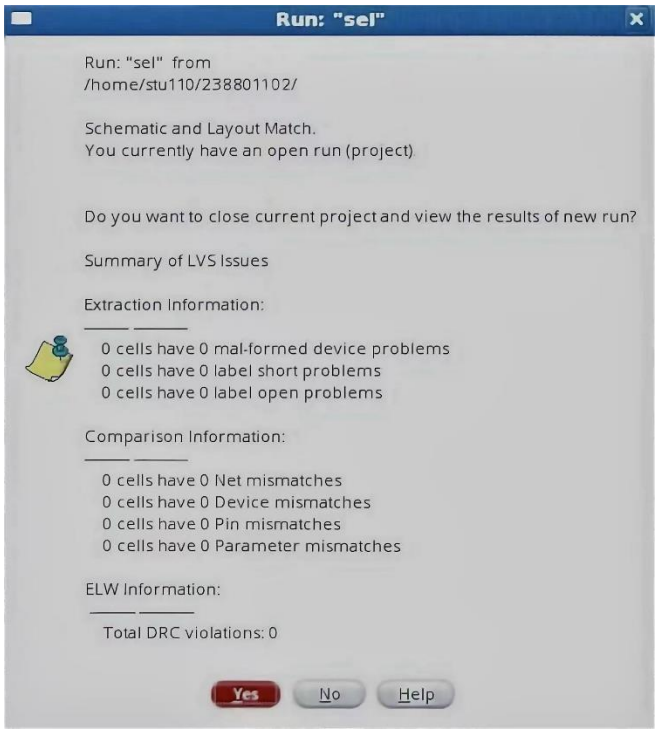
From the Virtuoso Layout Editor, **Assura → Run LVS** was selected. In the **Run Assura LVS** window, the schematic and layout cells from **lib7** were selected, and the technology settings were verified. Once all parameters were confirmed, the **OK** button was clicked to start the LVS comparison.



LVS Results

The LVS verification confirmed that the **layout matches the schematic** successfully. The comparison reported **zero net mismatches, zero device mismatches, zero pin mismatches, and zero parameter mismatches**. In addition, no malformed devices or connectivity problems were detected.

The successful LVS verification confirms that the physical layout correctly implements the intended circuit and is electrically identical to the original schematic, making the design ready for parasitic extraction and post-layout simulation.



Conclusion

In this laboratory assignment, the **SEL** circuit was successfully designed and implemented using the **Cadence Virtuoso** design environment with **gpdk045 (45 nm)** technology. The design was developed hierarchically by integrating **three NAND gate cells from lib2** and **one inverter cell from lib1** into the **lib7** library.

The complete design flow was carried out, including schematic creation, symbol generation, testbench implementation, circuit simulation, physical layout design, and verification. The layout was successfully verified using **Design Rule Check (DRC)** and **Layout Versus Schematic (LVS)**, confirming that the layout satisfies the fabrication rules and accurately matches the original schematic.

This laboratory provided practical experience with the complete CMOS IC design flow and demonstrated the importance of hierarchical design, layout verification, and design validation in modern VLSI circuit implementation.